

MUHAMMAD NAVEED

AI Agent Engineer | Forward-Deployed Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI Agent Engineer specializing in building AI systems that run reliably without constant supervision. Production track record: designed and operate five autonomous Slack/Google Workspace agents plus a live operations dashboard for a U.S. pest-control enterprise, handling real-time multi-step decision-making across 123 properties and 8+ field technicians, worth \$50K+/month in operational impact — including diagnosing and resolving a week-long silent production outage through systematic root-cause analysis. Deepened this production experience with 44 self-built, self-tested projects covering the full reliability stack modern agentic AI demands: stopping conditions and scheduled autonomy (Loop Engineering), permission guardrails and prompt-injection defense (Harness Engineering), shared memory across multi-agent systems (Graph Engineering), and evals-based testing to make "the checker said PASS" a defensible number (Trusting the Checker). Full-stack (Node.js, Python, FastAPI, Next.js) with production Claude/Anthropic API, OpenAI, and MCP integration experience. Open to US remote roles or relocation.

TECHNICAL SKILLS

Agentic AI	Autonomous agent design, loop engineering, guardrail/harness engineering, multi-agent shared memory, evals & golden-set testing, prompt-injection defense, MCP, LangChain, Agents SDK, OpenRouter
AI / LLM APIs	Anthropic API (Claude Code, production), OpenAI API, Gemini API — production integration at 10B+ tokens/month scale
Backend	Node.js, Python, FastAPI, Express, REST APIs, node-cron (production, 3+ years)
Frontend	Next.js, React, TypeScript, Tailwind CSS, HTML5/CSS3 (3+ years)
Integrations	Slack Bot & Events API, Google Sheets/Calendar/Gmail API, Webhooks, WhatsApp, Twilio
Cloud & DevOps	Railway (production), Docker, Kubernetes, Kafka, Dapr, CI/CD, GitHub, persistent volume storage, production log/health auditing
Databases	MongoDB, PostgreSQL, Sanity CMS
Architecture	Event-driven & spec-driven design, fuzzy matching/alias resolution, background precompute & caching, rate-limit-safe pipelines

AGENTIC AI ENGINEERING — SELF-BUILT PRACTICE PORTFOLIO (44 PROJECTS, 5 REPOS)

Beyond production work, completed a structured, hands-on curriculum in the engineering disciplines behind reliable AI agents — every project self-built, self-tested, and publicly documented on GitHub.

Loop Engineering (12)	Scheduled & event-driven agent loops, stopping conditions, maker-checker verification, self-healing "dreaming" loops that improve their own rules. github.com/NaveedTechLab/Loop-Engineering-Projects
Harness Engineering (8)	Permission guardrails, hooks, typed output validation, and a live prompt-injection attack & defense exercise. github.com/NaveedTechLab/Harness-Engineering-Projects
Graph Engineering (8)	Shared memory across multiple agents — typed claims with provenance, grounded fact-checking, entity resolution. github.com/NaveedTechLab/Graph-Engineering-Projects
Trusting the Checker (8)	Evals engineering — golden-set testing, judge calibration, CI regression gates, adversarial/injection test suites. github.com/NaveedTechLab/Trusting-the-Checker-Projects
Leaving the Laptop (8)	Production runtime patterns — headless deployment, unattended scheduling, portability drills, credential hardening. github.com/NaveedTechLab/Leaving-the-Laptop-Projects

PROFESSIONAL EXPERIENCE

AI Automation Engineer (Forward-Deployed) April 2026 – Present
Activus Capital Partners — Heat Wave Pest Control | Remote (Karachi-based)

Own five production agentic automation systems and a live operations dashboard for a multi-region pest control operator. Report directly to the Managing Partner, Operations Coordinator, and Operations Director.

Specialty Form Compliance Bot

- Automated daily monitoring of 100+ calendar events against Google Forms submissions across 2 U.S. regions, scaling to 123 properties via automatic header detection and property-scoped filtering
- Built a fuzzy-matching engine (compound-word decomposition, Levenshtein distance, alias resolution) to reconcile technician shorthand against property records

Notes Verification Bot

- Built real-time scheduling-notes verification triggered by technician Slack messages; scaled from single-region pilot to a full 8-technician, 123-property deployment with 15-minute escalation logic
- Diagnosed and fixed two production validation bugs (cross-property collision, cross-technician false-flagging) causing compliance false positives on shared multi-tech properties

Weekly Reporting Suite & Specialty Recommendation System

- Built and iteratively refined two automated weekly reports to exact stakeholder-specified formats across multiple feedback rounds
- Diagnosed a week-long silent production outage via a full read-only audit (307 cases, 61 properties); re-enabled with deduplication safeguards and a kill switch for safe rollout
- Redesigned per-service pricing logic to parse property-specific, day-conditional rules directly from source sheets, correcting pricing errors across multiple properties in one structural fix

Operations Dashboard (Live Web Application)

- Designed and built a password-protected, 6-section live dashboard (Express + vanilla JS) covering route timing, unit allocation, notes bot activity, and per-technician reporting
- Diagnosed a Google Sheets API rate-limit failure from 112+ concurrent reads; redesigned around background precompute and snapshot caching, cutting load time from 60–90s to instant
- Added CSV export and proactive disambiguation prompts for ambiguous technician input instead of silent failures

Engineering Practices

- Pre-flight verification and dry-run validation before every production matching-logic change and deploy
- Traced a week-long outage to a single misspelled environment variable via systematic hypothesis elimination; extended the health-check endpoint and audited 26 env vars to pre-empt recurrence
- Migrated all runtime data to persistent volume storage (zero data loss on deploy) and all credentials to company-owned accounts

Founder & Lead Developer

2025 – Present

NaveedTechLab (Digital Agency) | Karachi, Pakistan

- Built a digital agency targeting local Karachi businesses; secured first two clients (CCTV World Karachi and Ha-Aeen Dentistry)
- Produced 6-month digital transformation roadmaps, partner agreements, and pricing strategies; delivered end-to-end from scoping through workflow automation delivery

Freelance Full Stack Developer

2023 – Present

Fiverr & Direct Clients | Remote

- Delivered Shopify stores, business websites, custom dashboards, CMS-based sites, and automation workflows end-to-end across Fiverr and direct-client engagements, with a strong repeat-business rate

PROJECT & HACKATHON EXPERIENCE

All projects live and deployed — demos at naveedtechlab-portfolio.hf.space

- **Autonomous AI Marketing Agency** (Hackathon, 2024) — Live AI-powered agency handling autonomous campaign strategy, content generation, and client outreach
- **Personal AI Employee (Digital FTE)** (Hackathon, 2024) — Autonomous employee with watchers, Gmail/WhatsApp automation, an Obsidian memory system, and MCP tools for 24/7 task execution
- **Customer Success FTE (CRM Agent)** (Hackathon, 2024) — Multi-channel support agent across Gmail, WhatsApp, and web forms with a PostgreSQL ticket system and FastAPI backend
- **Cloud-Native Microservices Architecture** (2024) — Reusable agent skills and containerized microservices on an event-driven architecture using Docker, Kafka, Dapr, and Kubernetes

EDUCATION

Governor House Initiative — GenAI, Web3 & Metaverse Program

Diploma in Computer Information Technology (DCIT) · Certificate in Information Technology (CIT) | 2023 – Present

GIAIC Final Marathon — Agentic AI Engineering Track, Panaversity

Structured curriculum: Loop Engineering, Harness Engineering, Graph Engineering, Trusting the Checker, Leaving the Laptop | In Progress, 2026

LANGUAGES

English (Fluent) | Urdu (Native)